aggravate impaired vascular endothelium, causing vasospasms. It is interesting to note that the first child had seizures after platelet transfusion and now has a learning disability. Of note, an ultrasonographic examination of the current fetus had shown a −1.9 SD in the head circumference at 29 weeks, but follow-up scan showed a less significant lag. Although unproved, it is also possible that some earlier in utero event, i.e., hemorrhage, had taken place before therapy was initiated.

There is still much unknown regarding the pathophysiology of this disease and the effects of therapy, such as exogenous immunoglobulins and platelet transfusions, on the long-term outcome of these infants. Platelet transfusions would be the most reliable way to elevate the platelet count. However, they need to be performed with caution. Immunoglobulin treatment may be a better alternative, as our patient's platelet count was 90,000/mm^3 with weekly intravenous doses of immune globulin. Although studies of long-term immune globulin therapy in patients with immunodeficiency syndrome have failed to reveal any significant risks to date, the long-term consequences of maternal-fetal intravenous immune globulin therapy have yet to be defined.

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Spontaneous abortion and subsequent adverse birth outcomes

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OBJECTIVE: Our purpose was to evaluate the association between spontaneous abortion and subsequent adverse birth outcomes.
STUDY DESIGN: Washington State birth certificate records for 1984 to 1987 were used to examine the association between spontaneous abortion and adverse outcomes in the subsequent live birth. Adverse birth outcomes were examined for women with one spontaneous abortion before the index pregnancy (n = 2146) and for women with three or more prior spontaneous abortions and no other prior pregnancies (n = 638); compared with women with no prior spontaneous abortions (n = 3099). Logistic regression was used to estimate the relative risk associated with prior spontaneous abortion of each adverse outcome.
RESULTS: Women with three or more prior spontaneous abortions were at higher risk for delivery at <37 weeks' gestation (relative risk 1.5, 95% confidence interval 1.1 to 2.1), placenta previa (relative risk 6.0, 95% confidence interval 1.6 to 22.2), having membranes ruptured >24 hours (relative risk 1.8, 95% confidence interval 1.2 to 2.9), breech presentation (relative risk 2.4, 95% confidence interval 1.6 to 3.6), and having an infant with a congenital malformation (relative risk 1.8, 95% confidence interval 1.1 to 3.0).
CONCLUSION: These findings suggest that common causes may underlie recurrent spontaneous abortion and certain adverse birth outcomes. They may also help guide clinical management of pregnancies in women with a history of recurrent spontaneous abortions. (Am J Obstet Gynecol 1992;166:111-6.)

Key words: Spontaneous abortion, preterm delivery, congenital malformation

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Spontaneous abortion before the 20th week of gestational age is reported to occur in 15% to 20% of recognized pregnancies.[1, 2] Approximately 1% of women experience three or more spontaneous abortions before the first live birth.[1] In spite of the frequency of this problem, the subsequent birth outcomes of women with prior spontaneous abortions have not been extensively studied. Previous studies have suggested

that women with a history of spontaneous abortion are at increased risk for preterm delivery,[3-5] placenta previa,[6] and having an infant with intrauterine growth retardation (IUGR)[7, 8] or congenital malformation.[9, 10] These studies have been limited by small sample sizes[5-9] and a lack of consideration of possible confounding variables.[3-10]

The current study sought to examine the incidence of adverse birth outcomes among women with one prior spontaneous abortion and women with three or more prior spontaneous abortions, compared with the incidence among women with no prior spontaneous abortions. We used a large data base that includes information on maternal age, parity, gravidity, smoking status during pregnancy, race, and prenatal care.

Material and methods

Data were taken from Washington State birth certificate records for the years 1984 through 1987. Obstetric ward clerks are generally responsible for abstracting information from the medical chart into the birth certificate form. Handbooks describing procedures for birth certificate completion are provided by the State of Washington. Additional instruction is provided by each hospital. A state field representative makes periodic visits to review completion procedures. Most adverse birth outcomes are represented by a checklist format. The completeness of reporting birth complications on the Washington State birth certificate was the subject of a recent study.[11] With the checklist format, reporting was 80% or better for most birth complications (e.g., preeclampsia, breech presentation, and cesarean section). Congenital malformations were reported less frequently, as judged by hospital discharge diagnoses. Reporting rates for congenital malformations ranged from 32% for polydactyly to 57% for cleft lip and palate. The birth certificate defines spontaneous abortion as fetal loss occurring before 20 weeks' gestation; first- and second-trimester spontaneous abortions are not differentiated.

Two "exposure" groups were constructed. The group of women with one prior spontaneous abortion consisted of all women having a live singleton birth in 1987 who had one prior spontaneous abortion and no other prior pregnancies. A second group of women with recurrent spontaneous abortions consisted of all women having a live singleton birth in 1984 to 1987, who had three or more prior spontaenous abortions and no other prior pregnancies. Births ending in fetal death were not included because of their small numbers (<0.7% of all pregnancies >20 weeks' gestation).

The unexposed comparison group consisted of 3099 randomly selected women having a live singleton birth in 1987 with no pregnancies prior to the index pregnancy. Use of this primigravid group thus controlled for the potential confounding effects of parity.

The adverse birth outcomes of interest were: low birth weight (<2500 gm), IUGR, preterm birth (<37 weeks' gestational age, calculated from the first day of the last menstrual period), congenital malformations, prolonged rupture of membranes (>24 hours), breech presentation, low 1-minute (<6) and 5-minute (<7) Apgar scores, abruptio placentae, placenta previa, and preeclampsia. The definition of a low Apgar score was chosen to allow comparison of the results with those of a previous study.[10] Intrauterine growth retardation was defined as birth weight <10th percentile for gestational age as determined by last menstrual period. The 10th percentile of birth weight by menstrual weeks' gestation was taken from a widely cited nomogram and table of birth weights.[12]

Maximum likelihood logistic regression[13] was used to estimate the relative risk associated with prior spontaneous abortion for each adverse birth outcome while controlling for confounding variables.

Results

The characteristics of women with one prior spontaneous abortion, three or more prior spontaneous abortions, and the primagravid comparison group are shown in Table I. Women with three or more spontaneous abortions were older and more likely to smoke, compared with the primigravid group. Women with one prior spontaneous abortion were intermediate for these characteristics. The differences in the proportion of smokers among women with one prior spontaneous abortion and women with three or more prior spontaneous abortions, compared with the proportion among women with no prior spontaneous abortions, were highly significant ($p < 0.001$) after controlling for age. The differences by maternal race, marital status, urban residence (living in an incorporated area), and prenatal care were relatively small.

Information was missing on <3% of the outcome variables examined, except for gestational age, congenital malformations, and preeclampsia. Gestational age was missing for 7% of the group with one prior spontaneous abortion, 9% of the group with the three or more prior spontaneous abortions, and 9% of the primigravid group. The totals missing for those with malformations were 14%, 13%, and 16%, respectively. All three groups had 10% of the preeclampsia data missing.

The frequencies of adverse outcomes by group are presented in Table II. Differences were found for low birth weight, IUGR, preterm delivery, congenital malformation, placenta previa, prolonged rupture of membranes, breech presentation, 1-minute and 5-minute Apgar scores, and preeclampsia, but not for abruptio placentae.

Table III displays the relative risk estimates with 95% confidence intervals for the adverse outcomes in Table

Table I. Characteristics of women during their index pregnancy by number of prior spontaneous abortions

Characteristic	*Women with no prior pregnancies (n = 3099)*	*Women with one prior spontaneous abortion (n = 2146)*	*Women with three or more prior spontaneous abortions (n = 638)*
Maternal age			
Mean (yr)	23.9	25.0	26.3
<20 yr (%)	22.8	15.5	10.3
20-24 yr (%)	34.8	34.6	30.6
25-29 yr (%)	26.3	28.9	30.6
>30 yr (%)	16.0	20.9	28.5
Smoker (%)	19.7	28.1	36.5
White (%)	87.6	89.5	91.2
Married (%)	72.4	76.2	75.9
Urban residence (%)	51.6	50.2	54.2
First-trimester prenatal care (%)	77.6	81.3	79.7

Table II. Number and percent of selected adverse outcomes in the index pregnancy by number of prior spontaneous abortions

	No prior pregnancy		*One prior spontaneous abortion*		*Three or more prior spontaneous abortions*	
Outcome	*No.*	*%*	*No.*	*%*	*No.*	*%*
Low birth weight	141	4.6	124	5.8	60	9.5
IUGR	133	5.0	94	4.9	41	6.4
Length of gestation <37 wk	220	7.8	174	8.7	63	10.8
34-36 wk	162	5.7	130	6.5	36	6.2
30-33 wk	47	1.7	26	1.3	19	3.2
<30 wk	11	0.4	18	0.9	8	1.4
Any malformation (except ductus)	53	2.0	36	2.0	21	3.8
Placenta previa	6	0.2	7	0.3	8	1.3
Membranes ruptured >24 hr	79	2.6	52	2.4	33	5.2
Breech	83	2.7	75	3.5	43	6.7
1 min Apgar score <6	206	6.7	175	8.2	63	9.9
5 min Apgar score <7	64	2.1	36	1.7	22	3.5
Preeclampsia	189	6.8	116	6.0	26	4.5
Abruptio placentae	25	0.9	19	0.9	5	0.8

II. Maternal age, smoking status, race, marital status, place of residence, and prenatal care were evaluated for confounding effects on all outcomes. Gestational age was evaluated as a potential confounder for all outcomes except IUGR. Only those variables that resulted in a meaningful change in the estimated relative risk were adjusted for in the analysis.

After adjustment for maternal age and smoking status, women with three or more prior spontaneous abortions had twice the risk of having a low-birth-weight infant (relative risk = 2.0, 95% confidence interval = 1.4 to 2.8) compared with that of women with no prior pregnancies. This association appears to primarily reflect an increased risk of preterm birth among women with recurrent abortions (relative risk = 1.5, 95% confidence interval = 1.1 to 2.1), rather than an increased risk of IUGR (relative risk = 1.2, 95% confidence interval = 0.9 to 1.7). The estimated relative risk for prematurity among women with three or more prior spontaneous abortions increased with increasing prematurity, from 1.2 (95% confidence interval = 0.8 to 1.8) for mild prematurity (34 to 36 weeks' gestation) to 2.7 (95% confidence interval = 1.0 to 7.3) for severe prematurity (<30 weeks' gestation).

Women with recurrent spontaneous abortions had an increased risk of delivering an infant with a congenital malformation (relative risk = 1.8, 95% confidence interval = 1.1 to 3.0). Ductus arteriosus was not included in congenital malformations because of its strong association with prematurity, which was already known to be associated with prior spontaneous abortions. Analysis by individual type of malformation was not possible because of small numbers. Analysis by major organ systems (genitourinary, cardiac, central ner-

Table III. Estimated relative risks and 95% confidence intervals* for selected adverse outcomes in index pregnancy by number of prior spontaneous abortions

	One prior spontaneous abortion		*Three or more prior spontaneous abortions*	
Outcome	*Relative risk*	*95% Confidence interval*	*Relative risk*	*95% Confidence interval*
Low birth weight	1.2	0.9-1.6	2.0	1.4-2.8
Intrauterine growth retardation	0.9	0.7-1.2	1.2	0.8-1.7
Gestation <37 wk	1.2	0.9-1.4	1.5	1.1-2.1
34-36 wk	1.2	0.9-1.5	1.2	0.8-1.8
30-33 wk	0.8	0.5-1.4	2.4	1.4-4.3
<30 wk	2.2	1.0-4.7	2.7	1.0-7.3
Any malformation (except ductus)	1.0	0.6-1.5	1.8	1.1-3.0
Placenta previa	1.6	0.5-6.0	6.0	1.6-22.2
Rupture of membranes >24 hr	1.0	0.7-1.4	1.8	1.2-2.9
Breech presentation	1.3	0.9-1.8	2.4	1.6-3.6
1 min Apgar score <6	1.3	1.0-1.6	1.2	0.8-1.6
5 min Apgar score <7	0.9	0.6-1.4	0.6	0.3-1.6
Preeclampsia	1.0	0.8-1.3	0.8	0.5-1.3
Abruptio placentae	1.2	0.6-2.3	1.2	0.4-3.1

*Low birth weight, IUGR, gestational age, and preeclampsia were adjusted for maternal age and smoking status. Placenta previa, prolonged rupture of membranes, and breech presentation were adjusted for maternal age and gestational length. One- and 5-minute Apgar scores were adjusted for maternal age, smoking status, and gestational length.

vous system, and limb defects) did not show a significantly elevated relative risk of malformation within any organ system.

Women with three or more prior spontaneous abortions also had elevated risks of placenta previa (relative risk = 6.0, 95% confidence interval = 1.6 to 22.2), prolonged rupture of membranes (relative risk = 1.8, 95% confidence interval = 1.2 to 2.9), and breech presentation (relative risk = 2.4, 95% confidence interval = 1.6 to 3.6), after adjustment for maternal age and gestational length. Additional adjustment for other variables, including smoking and prenatal care, did not alter these estimates.

After adjustment for maternal age, smoking status, and gestational length, women with recurrent spontaneous abortions had a relative risk of 1.2 (95% confidence interval = 0.8 to 1.6) of having an infant with a low 1-minute Apgar score. Women with one prior spontaenous abortion had a relative risk of 1.3 (95% confidence interval = 1.0 to 1.6) of having an infant with a low 1-minute Apgar score. There was no evidence of an increased risk of preeclampsia or abruptio placentae in either group.

Comment

Investigating the association between prior spontaneous abortion and subsequent adverse birth outcomes is complicated by the interrelationship between parity, gravidity, maternal age, smoking, number of prior spontaneous abortions,[14] and the interrelationship between adverse outcomes such as low birth weight, IUGR, prematurity, and low Apgar scores. The current study first simplified the examination of prior spontaneous abortion by limiting the study to women with prior spontaneous abortions who had experienced no other prior pregnancies. Women with one prior spontaneous abortion and women with three or more prior spontaneous abortions were examined separately because previous studies have suggested that women with three or more prior spontaneous abortions may be at higher risk for subsequent adverse pregnancy outcomes than women with only one prior spontaneous abortion,[2, 4, 6] The finding that smoking and maternal age were positively associated with the number of prior spontaneous abortions was expected, as both smoking[15, 16] and advanced maternal age[14, 17] have been associated with spontaneous abortion and women who reported smoking in the index pregnancies would be likely to have smoked during previous pregnancies. In this data set, maternal race, marital status, and prenatal care were similar among women with and without prior spontaneous abortions, and adjustment for these variables did not change the associations reported.

In the current study the adjusted relative risk of IUGR was 1.2 (95% confidence interval = 0.8 to 1.7) in women with three or more prior spontaneous abortions. This result is in contrast to a previous study that reported an increased unadjusted risk for severe IUGR in term infants (calculated relative risk = 2.2, 95% confidence interval = 1.6 to 2.9) in women whose previous

pregnancy ended in a spontaneous abortion when compared with women whose previous pregnancy ended in a live birth.[7]

The current study found only a slightly elevated adjusted relative risk of 1.2 (95% confidence interval = 0.9 to 1.4) for preterm birth in women with one prior spontaneous abortion but an adjusted relative risk of 1.5 (95% confidence interval = 1.1 to 2.1) for women with three or more prior spontaneous abortions. These results are similar to results reported from previous studies.[3, 4, 10]

Previous studies have suggested that prior spontaneous abortion is associated with an increased incidence of congenital malformation in a subsequent pregnancy.[9, 10] The rate of congenital malformation reported in the current study is less than that reported in previous studies.[10, 18] This difference probably reflects an underreporting of congenital malformations on the birth certificate. The current study found no evidence of an increased risk of congenital malformation in infants of women with one prior spontaneous abortion. However, women with three or more prior spontaneous abortions had a relative risk of 1.8 (95% confidence interval = 1.1 to 3.0) of having an infant with a congenital malformation. A larger study would be needed to evaluate the association between prior spontaneous abortions and specific malformation types.

A strong association was found in the current study between three or more prior spontaneous abortions and placenta previa (relative risk = 6.0, 95% confidence interval = 1.6 to 22.2) after adjusting for maternal age, smoking status, and gestational length. A previous study, which examined the combined risk of placenta previa and abruptio placentae in women with one prior spontaneous abortion, reported data that gave a calculated reltive risk of 2.3 (95% confidence interval = 0.7 to 7.9).[10] A more recent case-control study of 80 cases of placenta previa found calculated odds ratios of 1.5 for one prior spontaneous abortion (95% confidence interval = 0.5 to 4.1) and 4.9 (95% confidence interval = 0.3 to 88.4) for three or more prior spontaneous abortions.[6] The suggested explanation for this association was that curettage, commonly performed after a spontaneous abortion, may be a risk factor for subsequent placenta previa.

The current study found a moderate association between prolonged rupture of membranes and three or more prior spontaneous abortions, which was independent of gestational age and early prenatal care. This finding has not been previously reported. It is possible that this result reflects an association between delay in seeking medical care at the time of membrane rupture and other, unmeasured factors related to spontaneous abortion.

Another, apparently new finding is an elevated relative risk for breech presentation among women with three or more prior spontaneous abortions (relative risk = 2.4, 95% confidence interval = 1.6 to 3.6) after adjustment for maternal age, smoking status, and gestational length. The explanation for this finding is not apparent, although it could reflect underlying morphologic uterine abnormalities that predispose to both spontaneous abortion and breech presentation.

Women with one prior spontaneous abortion had a relative risk of 1.3 of having an infant with a 1-minute Apgar score <6, after adjustment for smoking, maternal age, and gestational age. This is less than the relative risk of 1.8 (95% confidence interval = 1.1 to 2.9) calculated from data reported in a previous study for a 1-minute Apgar score <6 among women with one prior spontaneous abortion.[10]

There are a number of limitations in the current study that should be noted. The use of birth certificate data to ascertain prior spontaneous abortion undoubtedly results in some misclassification of the number of prior spontaneous abortions. Because the information on prior spontaneous abortion is taken from the medical record, where it is usually recorded before the birth outcome, it is unlikely that prior spontaneous abortions were preferentially recorded for women with adverse outcomes. As long as such misclassification is not related to the outcomes of interest, the reported relative risk would be a conservative estimate of the true relative risk.

It is also possible that the associations reported may be the result of an unmeasured external variable, such as poor nutrition or alcohol or drug abuse, rather than a shared biologic mechanism. Socioeconomic status was not adjusted in the current study; however, mother's household income did not confound the associations in a pilot analysis with the same data. Moreover, adjustment for race, marital status, and first-trimester prenatal care, all of which would be expected to be related to socioeconomic status, did not affect any of the associations reported.

It is possible that the association between the adverse outcomes reported and three or more prior spontaneous abortions actually reflects an association with second-trimester spontaneous abortions,[19] which would be more common among women with three or more prior spontaneous abortions, rather than an association with the number of prior spontaneous abortions per se. This possibility could not be addressed in the current study because the data did not distinguish between first-trimester and second-trimester spontaneous abortions.

Multiple, nonexclusive causes have been suggested for spontaneous abortion, including chromosomal aberrations,[2, 20] maternal immunologic[21, 22] or hormonal[23] abnormalties, infection,[24] incompetent cervix,[25] and uterine malformations.[2] One or more of these factors may also result in an adverse birth outcome, creating an association between prior spontaneous abortion and

specific adverse birth outcomes. IUGR, for example, could plausibly result from immunologic, implantation, infection, or hormonal defects. Prematurity and IUGR have been associated with both maternal infection[26] and immunologic abnormalities.[21] Congenital malformations, while possibly related to chromosomal aberrations, may also be the result of infection or of immunologic or hormone factors.[27] In spite of these complexities in the interpretation of the relation between prior spontaneous abortion and subsequent adverse birth outcomes, it is hoped that the results of the current study will provide clues to unraveling the etiology of spontaneous abortion.

The findings from the current study suggest that women whose first pregnancy ended in a spontaneous abortion are at little, if any, increased risk for an adverse outcome in a subsequent live-birth pregnancy. However, women with recurrent (three or more) spontaneous abortions and no other prior pregnancies are at increased risk for several adverse outcomes, including placenta previa, preterm delivery, and breech presentation. These results may be useful in guiding the clinical management of pregnancies in women with recurrent abortions.

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